Kori Vernon

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Experience

Aug 2024 – Present **Electronic Sales Trader, Associate**, *Institutional Equities*, Electronic Trading (MSET), Morgan Stanley

- Cover retail aggregators, market makers, exchanges, and institutional broker-dealers in US equities and options, representing ~$30B in executed value.
- Leading the product build-out of fractional share trading for broker-dealer clients, defining requirements and coordinating [technology, product, and compliance] teams across [scope, e.g. order handling, allocation, and client onboarding].
- Advise clients on algorithm selection and parameter tuning (VWAP, TWAP, POV, smart order routing), running post-trade TCA reviews to diagnose slippage and recommend strategy changes.
- Built Python tools automating client TCA reporting, desk workflows, and operational tasks.

Nov 2022 – Jul 2024 **Data Engineer, Associate**, *Institutional Securities Technology*, Internal Trading and Reporting Applications, Morgan Stanley

- Built and deployed a seasonal ARIMA model forecasting trade execution volumes, recalculated every 15 minutes and typically within 3–5% of realized volume during core trading hours, integrated end to end with a Python/SQL (DB2) backend and UI.
- Developed a Data Quality Rule Suggestion Engine that profiles datasets and proposes validation rules, enabled across the entire platform supporting the firm's internal equities trading and reporting.
- Built data pipelines and APIs for trading and reporting apps, applying ML to improve data quality and modernize systems.

Feb 2022 – Nov 2022 **Automated Risk Trading Production Engineer, Analyst**, *Reliability and Production Engineering*, Automated Market Making (AMM), Morgan Stanley

- Provided DevOps and production support for the AMM business, which delivered billions of quotes per day across 15 US exchanges, triaging live trading incidents with traders and developers.
- Built real-time monitoring of key business indicators across the AMM trading platform.
- Enhanced production systems and automated operational processes to improve efficiency and keep trading running smoothly.
- Built a report aggregation tool and dashboard for AMM DevOps to load, query, export, and purge production data, speeding up issue investigation and resolution.

Projects

Aug 2026 – Present **PomTrader**, *Personal Trading Platform*, Python

- Built and operate a platform that manages personal brokerage accounts end to end, integrated with the E*Trade broker API for positions, market data, and execution.
- Run a live systematic strategy trading leveraged Bull/Bear ETFs on combined MACD and RSI signals, while developing and tuning several additional strategies in paper trading.
- Designed a hedging engine that recommends ETF, options, and beta hedges sized to portfolio exposures.

Aug 2026 – Present **Sports Event Market Maker**, *Personal Project*, Python

- Built an automated market maker quoting two-sided markets on Kalshi and Polymarket across all sports markets, pricing fair value from a proprietary model.
- Manage inventory with per-event exposure limits, skewing quotes on net position to control risk and to deliberately accumulate inventory into games where the proprietary model has high confidence.

Apr 2023 – Present **Equities Monitor (EMon)**, *Personal Project*, Python

- Designed the paper-traded Intraday Volatility Outlier Mean Reversion Options (IVOMRO) strategy, which scans ~6,900 US equities every 15 minutes, flags names whose move exceeds 3-month realized vol, then prices candidate options with Black-Scholes using realized vol against market implied vol to open simulated positions, tracking P&L and exiting losers on predefined risk rules.
- Built the Earnings Equity Options (EEO) strategy, a daily scheduled screen ranking earnings plays by historical post-earnings moves.

Education

Aug 2018 – Dec 2021 **New York University**, *Tandon School of Engineering*, B.S. Computer Science, Minor in Mathematics

Skills & Licenses

Trading Market structure, options Greeks, volatility estimation, market making (two-sided quoting, inventory management, quote skewing), algorithmic execution (VWAP, TWAP, POV), smart order routing, TCA

Languages Python, SQL; working knowledge of C++, Java, kdb+/q

Licenses SIE, Series 7, Series 57, Series 63

Interests Powerlifting, Paintball, Cars, Cooking, Soca, Volunteering & Mentorship (America Needs You)